Voi / Portfolio Lab

Hub & Book investment evidence

Named report case | Model 7.2.1 | Frozen 2026-11-02. Displayed numbers are rounded; downloads retain exact values.

Illustrative schedule: first operating month October 2026; setup September 2026. Calendar labels do not change the model timing.

Independent student analysis. Not affiliated with, endorsed by or commissioned by Voi.

The proposed priority is a Hub & Book pilot, followed by expansion only if customer response and delivery economics support the investment. This is a proposed strategic allocation of resources; the simulator continues to show competing designs and can recommend a different economic choice.

Table B2. Three nearest profitability boundaries

Input	Current	Zero-value boundary
Booking fee	35.0 kr	27.3 kr to 44.9 kr
Placement share per booking	20.0%	25.6%
Field minutes per placement	8.0 minutes	11.1 minutes

One input at a time. These are zero-value boundaries, separate from value / peak cash at least 1, payback within 24 months and 95% fulfilment.

This is a normative capital-allocation policy, chosen independently of the case result; it is not an observed Voi hurdle or an empirically estimated optimum. The value hurdle therefore follows the selected path's cash requirement: NOK 623,000 in the improvement case. That case meets both targets; the reference case does not meet them.

The improvement objective is 20% dedicated additional placement while maintaining 95% fulfilment. The scope is an estimated 124,822 active Oslo customers, phased expansion from month 10, and continued operating rights. These figures are for the full pilot-and-expansion programme, not a small pilot alone.

Table B3. Three delivery scenarios

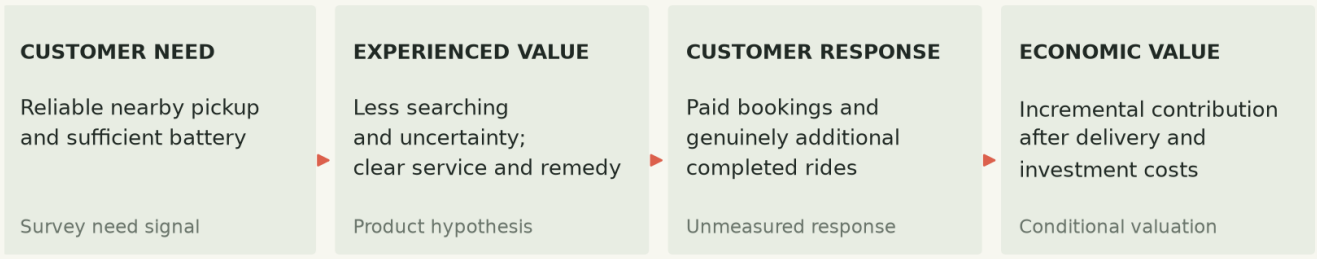
Assuming expansion	Placement	Economic value	Payback	Max cash need
Improvement objective	20%	+NOK 0.95m	June 2028 (month 21)	NOK 623,000
Reference	25%	+NOK 0.11m	June 2029 (month 33)	NOK 663,000
Higher placement stress	30%	-NOK 0.85m	Not within 36 months	NOK 906,000

At 30% placement, expanding despite a failed gate gives -NOK 0.85m. Its NOK 906,000 maximum cash need is the month-36 deficit and is still growing at the horizon; it is not an established lifetime peak.

36-month horizon. Assumed annual discount 10%. No terminal value. Default relationship uplifts are zero. Resource costs and cash are distinguished; an unspecified funding ceiling does not confirm affordability.

Customer value and the operating argument

Figure B3. Customer need to portfolio value



Customer need supports relevance. It does not establish demand for this exact booking product or a treatment effect.

Convenience Seekers rated adequate battery at 6.68/7 and nearby availability at 6.23/7 in the 72-person convenience sample. These descriptive need signals motivate a reliability proposition; they do not estimate booking adoption, population shares or willingness to pay.

Assumption	Planning value	Argument and evidence needed
Hub concentration	20% additional placement; 40% eligible reach	Select approved hubs and departure windows where usable inventory is already available after the 23:00-05:00 closure. Reduced placement must not simply remove customers. Measure requests, failed searches and extra crew jobs.
Service reliability	95% fulfilment	Verify a usable, sufficiently charged vehicle at the promised location and time. Record failures and remedies; measure reliability jointly with placement and reach.
Price and response	NOK 35 fee; 8 bookings per adopter/month	A convenience need does not establish willingness to pay. Test fee conversion and repeat booking among eligible customers; include non-users in the economic evaluation.
Demand response	15% frequent / 30% subscriber adoption at the entered fee	Adoption is price-adjusted from a reference fee; both rates and the response coefficient remain assumptions. Record conversion at tested prices and compare assigned cohorts.
Scale and unit economics	124,822 estimated active customers; 23.06 net NOK/ride	Oslo scale and revenue allocation are constructed estimates. The survey is not representative. Validate city customer counts, segment reach and marginal costs before committing to scale.
Internal resource cost	100% of internal office-hour cost treated as new cash	Keep the full economic opportunity cost. Finance must distinguish hiring, contractor or replacement cash expenditure from existing salaried capacity.

The hypothesis must work jointly: choosing better-supplied hubs can lower placement cost, but a narrower footprint may also reduce customer relevance and demand. The model does not estimate that joint trade-off automatically.

A worked delivery unit

A dedicated placement costs NOK 73 under the entered 8 minutes, NOK 400/hour and NOK 20/job transport assumptions. At 20% placement, this is NOK 15 per booking attempt before overtime. Fee revenue is net of VAT and conditional on fulfilment; support, compensation, displacement and payment costs also apply.

Investment choices and uncertainty

Each investment faces the same customers, external demand path and no-offer counterfactual. The two implementation paths are shown separately; they must not be ranked on mixed bases.

Table B1. Five investment choices

Investment	Pilot value	Direct value	Pilot max cash	Value / cash
No investment	NOK 0.00m	NOK 0.00m	NOK 0	-
Hub & Book	+NOK 0.95m	+NOK 1.41m	NOK 623,000	1.53
Voi Together	-NOK 0.13m	-NOK 0.02m	NOK 375,000	-0.36
Both current designs	+NOK 0.97m	+NOK 1.54m	NOK 930,000	1.04
Hub & Book + coordination + split payment	+NOK 1.07m	+NOK 1.72m	NOK 987,000	1.09

The highest-value pilot-then-expansion design is Hub & Book + coordination + split payment at +NOK 1.07m; the same design gives +NOK 1.72m on direct rollout.

The selected pilot plan ranks the proposed investment; direct rollout is shown alongside on the same baseline. Capital efficiency informs Hub-first without changing the economic ranking.

Feasibility and funding

For the selected improvement path: Funding ceiling unspecified. Launch cash is NOK 410,000; peak cash is NOK 623,000. Launch-budget flags and the hard total-cash ceiling are separate checks. Neither positive value nor an affordable launch guarantees that the whole path can be financed.

Paired uncertainty on the pilot path

Candidate	5th percentile	Median	95th percentile	Below zero
No investment	NOK 0.00m	NOK 0.00m	NOK 0.00m	0.0%
Hub & Book + coordination + split payment	-NOK 1.61m	+NOK 0.26m	+NOK 2.60m	41.6%
Hub & Book	-NOK 1.57m	+NOK 0.20m	+NOK 2.49m	43.0%
Voi Together	-NOK 0.47m	-NOK 0.23m	+NOK 0.23m	82.0%
Both current designs	-NOK 1.69m	+NOK 0.15m	+NOK 2.48m	44.6%

3,000 paired draws; seed 2026. The selected booking option and no investment are included. Values follow the pilot path. Shares describe assumed combinations, not measured chances of success. Range symmetry and dependence are modelling choices; changed ranges are not improved product evidence.

The target does not price unknown unknowns

Reassess the case when operating permissions, rival pricing, technology or customer behaviour changes materially. Switching costs, satisfaction and actual repeat use are distinct. Current economics does not establish durable protection from imitation.

Staging, joint assumptions and scale

For Hub & Book, direct rollout gives +NOK 1.41m against +NOK 0.95m for pilot then expansion: staging forgoes NOK 0.45m if the improvement assumptions hold. These are conditional full-programme values, not pilot-only outcomes. A failed gate requires stop or redesign, not automatic expansion.

Placement	Direct value	Pilot + expand	Stop at gate	Stop max cash
20%	+NOK 1.41m	+NOK 0.95m	-NOK 0.51m	NOK 512,000
25%	+NOK 0.37m	+NOK 0.11m	-NOK 0.53m	NOK 538,000
30%	-NOK 0.80m	-NOK 0.85m	-NOK 0.56m	NOK 564,000
35%	-NOK 2.32m	-NOK 2.08m	-NOK 0.58m	NOK 590,000

The stop path ends service after the month-9 gate and includes the following wind-down month. Modelled loss runs from -NOK 0.51m to -NOK 0.58m across the tested placement shares. This protection depends on following the gate; continuing after failure is a different scenario.

Reach and placement must work jointly

Eligible reach	20% placement value	25% placement value
40%	+NOK 0.95m	+NOK 0.11m
36%	+NOK 0.72m	-NOK 0.04m

Hubs with existing evening inventory can reduce extra placement, but a smaller footprint may reduce access and demand. The 23:00-05:00 closure constrains pickup windows. The model does not estimate that joint trade-off automatically.

Validate the eligible segment before launch

Convenience share	Pilot value	Value / cash	Policy met
25%	+NOK 0.58m	0.91	No
20%	+NOK 0.23m	0.35	No

Sample shares are scenario weights applied to the market estimate; they are not measured population shares. Segment-share scenarios also redistribute the remaining portfolio composition.

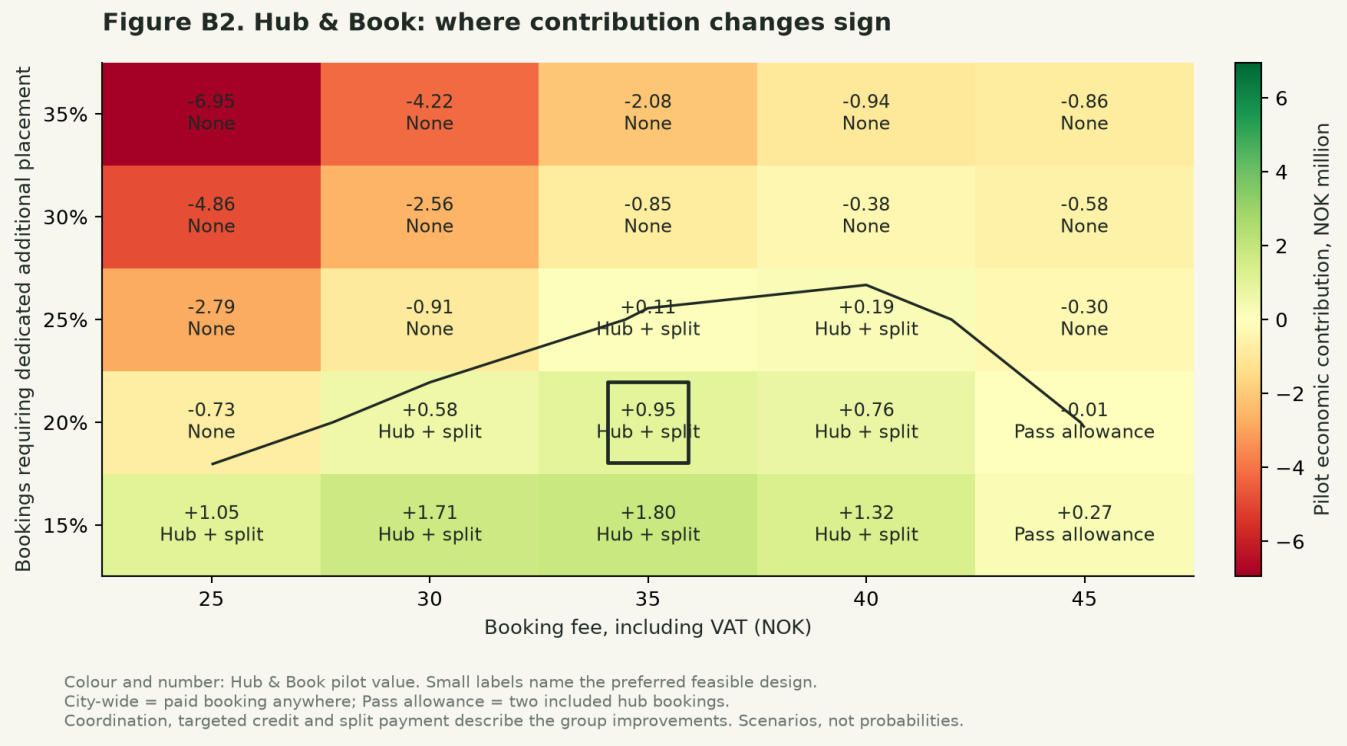
Horizon sensitivity: 24 months: +NOK 0.25m; 36 months: +NOK 0.95m; 48 months: +NOK 1.60m. No terminal value is credited. The 36-month horizon is a conditional medium-term appraisal, not a guaranteed operating entitlement.

The dated 2024 market-input stress gives +NOK 0.00m and september 2029 (month 36). Continued operating rights are a condition; the current permit ends in March 2027 before the June 2027 gate.

Across 3,000 assumed paired combinations, Hub & Book has median +NOK 0.20m, 43.0% below zero and 35.1% meeting both capital-policy targets. These shares describe tested assumptions, not observed probabilities.

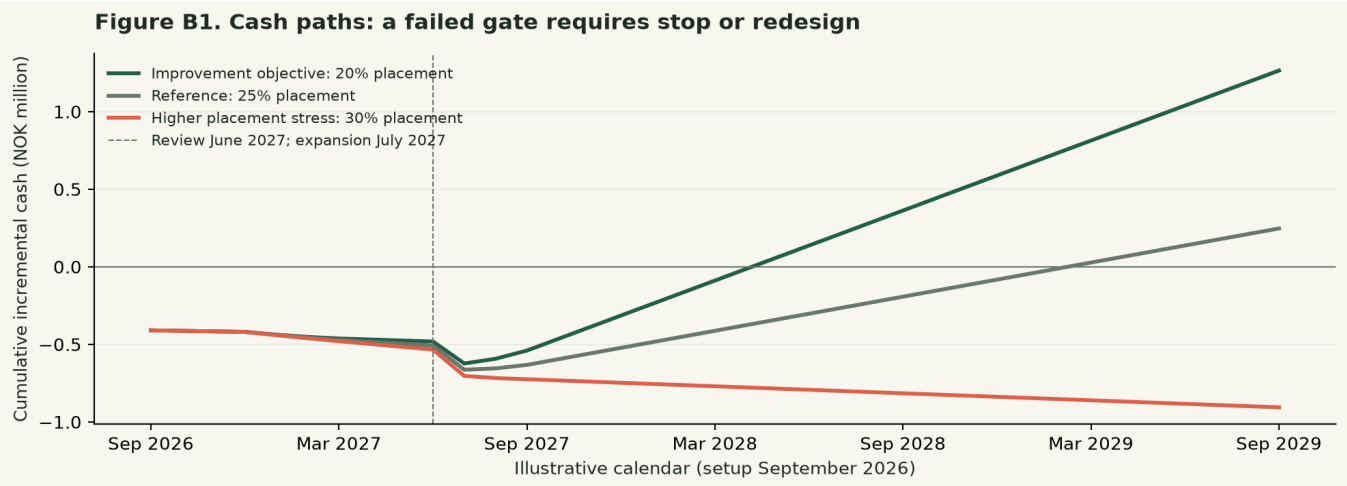
Decision boundary and cash

Figure B2. Decision boundaries



Values and the zero line follow the selected Hub & Book pilot path. Small labels identify the preferred feasible pilot design at the same fee and placement assumptions. The report choice and the highest-value catalogue design remain distinct.

Figure B1. Monthly incremental cash



The improvement, reference and stress cash curves come from their frozen monthly rows. Economic value retains full internal resource cost. Additional internal cash share is 100%; a different staffing plan changes cash needs without removing the opportunity cost.

Pilot delivery and decision protocol

Milestone	Model timing	Interpretation
First launch	January 2027	Benefits start after required capabilities are ready.
First cohort completes ramp	April 2027	Later expansion cohorts each have their own ramp.
Operational review	After June 2027	Expansion starts in month 10 only if the review supports it.
Retention evidence window	Ends October 2027	Six complete post-ramp months; any claim needs concurrent control through its window.
Control release	After June 2027	15% of the market is held as control until release.

Economic outcome and customer guardrails

Measure incremental contribution per assigned eligible customer, including non-bookers, against a concurrent control. Use stable hub-zone assignment, record spillover and analyse at the assignment level. Pre-specify fulfilment, battery sufficiency, customer effort, complaints and refund guardrails. Measure fee conversion, repeat booking, placement minutes and displaced trips together.

Track discounted economic value at least 1 times the selected path's maximum cash need and payback by month 24. Approve expansion only if future incremental contribution exceeds remaining avoidable costs, service reliability and operating rights hold, and cash/capacity constraints are satisfied. A failed gate means stop or redesign. Exclude sunk setup from continuation economics. The month-9 gate does not validate retention; no such uplift is credited here.

Owner	Accountability
Country manager / programme owner	Sponsor, cross-functional authority, review and escalation.
Operations	Approved hub locations; qualified supply; capacity, crew jobs and fulfilment records.
Product and Commercial	Service promise, booking flow, price tests, terms and remedies.
Analytics	Assignment, contamination, assigned-customer outcomes and updated appraisal.
Finance	Economic cost versus additional cash, funding, and continuation assessment.

Study precision is not assumed

Required inputs include eligible participants, independent clusters, cluster size, ICC, outcome variance, attrition and minimum worthwhile effect. Until these are supplied, there is no sample-size or detectable-effect claim. Few clusters and interference need specialist design validation.

A separate five-minute reader test checks usability. Automated browser and numerical checks do not establish usability or empirical validity.

Methods, report use and references

Use in the report

The core report must explain the customer-centred diagnosis, financial goals, strategic priority, budget and implementation conditions without requiring this simulator. Use the downloadable report mapping for section-by-section guidance; this supplementary evidence supports the argument and does not replace a required section.

Methods and limits

Economic contribution is the discounted difference between each offer and the same no-offer world, less resource setup. All configurations use 36 months and 10% annual discount. Monthly cash instead uses incremental cash expenditure. The active-customer scale, Oslo revenue allocation, adoption, additionality and operating costs remain assumptions or derived proxies. Usage states do not measure relationship quality.

The public data audit did not establish a compatible historical Voi scooter-trip series. Prophet remains off. Imported demand factors must affect the offer and counterfactual consistently. A forecasting model would require historical validation against a seasonal baseline.

Complete source and assumption register, monthly results, methods and pilot protocol accompany this PDF. Configuration cc19695f4f1a48599f5bccc5995f3d76110c861d4ca54f517ae45911868e9331. Figures and tables use the same frozen inputs. The core report must remain understandable without opening this supplementary pack.

Historical paper reference: unchanged v7.2.0 archive, https://kingen560.github.io/voi-portfolio-engine/archive/v7.2/paper_evidence_pack/. Archive configuration SHA256 cc19695f4f1a48599f5bccc5995f3d76110c861d4ca54f517ae45911868e9331. This PDF is the current 7.2.1 evidence pack.

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